

PROMPT-DRIVEN DETECTION OF URBAN TRAFFIC CONGESTION PATTERNS USING REAL-TIME GEOSPATIAL DATA

RESEARCH QUESTION

Can prompt engineering and lightweight AI models help detect and interpret traffic congestion patterns in urban areas?

INTRODUCTION

For this lab, I chose an area of interest in the transportation industry to which AI can be applied: identifying clusters of congested roadways in a downtown city block and generating insights from the data. Traffic congestion is an issue that impacts public health, safety, and sustainability in urban areas across the United States. Traditionally, diagnosing congestion on roadways requires the use of cumbersome traffic models, large-scale infrastructure, and counts taken in the field. This project explores the question of whether lightweight AI and LLM-based prompt strategies can be employed as low-cost, accessible, scalable tools for interpreting congestion metrics.

To test the research question, this project will first identify a small section of downtown San Francisco, California that comprises approximately 300 road segments of varying road types. These segments will incorporate data enrichment from OpenStreetMap and real-time traffic information accessed via the HERE Traffic API. By using accurate, up to date data from the API, we will integrate attribution such as Jam Factor, Speed, and Free Flow speed to determine where congestion is occurring, and at what rate. We will apply clustering algorithms such as K-Means and DBSCAN to these segments to identify congestion clusters and label roads as “congested” or “non-congested” based on cluster membership. Finally, the cluster-labeled geospatial dataset showing congestion zones will be fed to an LLM to help understand spatial results and provide recommendations to reduce congestion.

DATA COLLECTION & PROCESSING

The goal of assembling the processed data for this lab was to create a small scale, time relevant dataset that reflects urban mobility patterns in downtown San Francisco. Two primary data sources were utilized: OpenStreetMap and HERE Traffic API.

OpenStreetMap (OSM) is a free, crowd-sourced map database that represents a global road infrastructure network updated by volunteers online. Outside of proprietary sources such as Google Maps and Apple Maps, OSM is considered one of the most complete and accurate depictions of existing roadway features worldwide.

The HERE Traffic API, an online service offered by HERE, provides real-time traffic flow and incident data in JSON format. This service is available in a free tier with a maximum of 5,000 monthly transactions. Several attributions for roadway segments are ingested from the API to help determine congestion status by the machine learning algorithms in this project:

- **Free Flow:** the reference speed (in meters per second) along the roadway when no traffic is present.

- **Speed:** the expected speed (in meters per second) along the roadway; will not exceed the legal speed limit.
- **Jam Factor:** the amount of traffic on a roadway, the value ranging from 0.0 to 10.0 indicating quality of travel.

The steps taken to convert raw data inputs from these sources into spatial and temporal features suitable for ML modeling and LLM analysis are as follows:

1. *Study Area Definition:*

First, a small bounding box was defined to represent the downtown San Francisco area using latitude and longitude coordinates. The purpose of this initial process was to limit the geographic scope down to a manageable dataset of ~300 road segments for the project to use efficiently.

2. *Road Network Extraction:*

The OSM Python library, osmnx, was used to download the road network within the bounding box, visualize it as a GeoDataFrame, and prepare the geometry for analysis.

3. *Real-Time Traffic Data Collection via the HERE API:*

The HERE Traffic API was queried via the v7 endpoint using the bounding box. Attributes were extracted to include values representing current speed (m/s), free flow speed (m/s), jam factor (0-10 score) and segment geometry (road network). The JSON was converted into a GeoDataFrame of road segments.

4. *Enrich data with OSM Road Classifications:*

To add another parameter whereby we can analyze why congestion is occurring where it does, road segments were enriched with road classification attribution from OSM. The road network was first downloaded using osmnx, then matched using spatial joins to the HERE road segments. Each HERE segment is then given an OSM-derived highway type, such as tertiary, secondary, and residential roadway classifications.

5. *Unsupervised Learning Feature Preparation:*

Several changes needed to be made to the dataset for Machine Learning ingestion. The key features that will be used by the clustering algorithms are Jam Factor, Highway Classification, Coordinates, and Speed Ratio.

Jam Factor is a key component, and values can be used as-is – this feature will be weighted highly. Highway types are encoded to integer for use by the models. Coordinates, available as X/Y values extracted from road geometries, are projected to Web Mercator. Finally, we derive Speed Ratio from the dataset by calculating the following equation:

$$\text{Speed Ratio} = \text{Free Flow Speed} / \text{Current Speed}$$

Unlike Jam Factor, Speed Ratio is normalized across road types and is intuitive for interpretation by K-Means and distance-based clustering. A Speed Ratio of 1.0 would indicate no congestion as traffic flows normally, while a Speed Ratio of 0.5 would indicate that traffic is moving at 50% of expected speeds.

Additionally, manual weighting is used to emphasize congestion related features before clustering: Speed Ratio x 10, Jam Factor x 5, Highway Classification x 1.

METHODOLOGY

This project aimed to utilize a geospatial machine learning workflow to identify and interpret traffic congestion patterns using real-time mobility data. Integrating data from the HERE Traffic API and OpenStreetMap allows us to leverage spatial attributes, derive traffic metrics, and employ unsupervised clustering to detect groups of congested roadways across a defined urban area in San Francisco. Beyond the data collection and processing phases, the methodology process involves three stages: clustering, data aggregation and summarization, and prompt engineering.

CLUSTERING TECHNIQUES

An unsupervised clustering method was used to categorize the road network based on congestion factors. The project initially experimented with the use of DBSCAN (Density Based Spatial Clustering of Applications with Noise). Since DBSCAN is widely used and popular for its ability to isolate outlier points based on spatial components, this made it a suitable candidate for identifying congested roadways without a predefined number of clusters.

The DBSCAN clustering was run using centroid coordinates for each road segment along with the congestion related parameters defined in the data processing phase: Jam Factor and Highway classification. After reprojecting the dataset to a planar coordinate system, the eps and minimum samples factors were changed iteratively to output a viable dataset. The following code snippet showcases the core logic of the DBSCAN clustering, wherein the spatial projection, feature scaling, and DBSCAN application occur:

```
# Reproject to meters
df_proj = df.to_crs(epsg=3857)
df_proj["x"] = df_proj.geometry.centroid.x
df_proj["y"] = df_proj.geometry.centroid.y

# Prepare features
df_proj["jamFactor_scaled"] = df_proj["jamFactor"] *
2.0
df_proj["highway_encoded_scaled"] =
df_proj["highway_encoded"] * 1.5

X = df_proj[["x", "y", "jamFactor",
"highway_encoded"]].values
X_scaled = StandardScaler().fit_transform(X)

# Apply DBSCAN
db = DBSCAN(eps=eps,
min_samples=min_samples).fit(X_scaled)
df["dbscan_cluster"] = db.labels_
```

CODE SAMPLE: DBSCAN CLUSTERING ATTEMPT

The result from this attempt indicates that DBSCAN was ineffective at producing meaningful clusters within the context of the input dataset. All segments were assigned to a single dominant cluster of 0: therefore, cluster definition seems to be driven by spatial proximity, instead of congestion intensity resulting from Jam Factor or Highway type (Table 1).

dbscan_cluster	Segment Count
0	359

TABLE 1: DBSCAN OUTPUT

We can interpret from this data that a lack of dense spatial clusters with higher, more consistent Jam Factors has affected the model's ability to group localized clusters. Due to these limitations, DBSCAN was deemed unsuitable for the structure of the traffic data and the key features used to define congestion.

The project then pivoted to K-Means clustering, which was more effective at segmenting roadways based on weighted parameters such as Jam Factor, Speed Ratio, Highway classification, and spatial location. K-Means clustering was chosen foremost for its simplicity and interpretability. The feature set used for clustering was normalized using StandardScaler to produce consistent scaling, and the clustering process was run on datasets that were collected live via the NEAR Traffic API during both peak (Monday 5pm PST) and off-peak (Sunday 5pm PST) hours.

For this process, Jam Factor and Speed Ratio were weighted more heavily than spatial characteristics to prioritize the real-time traffic data while also including contextual separation between roadways. In the following code sample, we can see logic behind the K-Means clustering model, with engineered features, encoded highway types, applied weights to important features, and K-Means utilization:

```
# Handle division and limit ratio
df["speed_ratio"] = df["currentSpeed"] / df["freeFlowSpeed"]
df["speed_ratio"] = df["speed_ratio"].clip(upper=1.0) # cap at 1.0

# Encode highway types
highway_types = df["highway"].astype(str).unique()
highway_map = {h: i for i, h in enumerate(sorted(highway_types))}
df["highway_encoded"] = df["highway"].astype(str).map(highway_map)

# Weight features
df_proj["jamFactor_weighted"] = df_proj["jamFactor"] * 5.0
df_proj["speed_ratio_weighted"] = df_proj["speed_ratio"] * 10.0
df_proj["highway_encoded_weighted"] = df_proj["highway_encoded"] * 1.0

# Assemble feature matrix
X = df_proj[["x", "y", "jamFactor_weighted", "speed_ratio_weighted",
"highway_encoded_weighted"]].values
X_scaled = StandardScaler().fit_transform(X)
# Apply KMeans
kmeans = KMeans(n_clusters=n_clusters, random_state=42, n_init="auto")
df["kmeans_cluster"] = kmeans.fit_predict(X_scaled)
```

CODE SAMPLE 2: K-MEANS CLUSTERING ATTEMPT

The K-Means clustering method produced clear separation between congested, moderate, and free-flowing traffic segments. Cluster results were evaluated based on the separation of average jam factors, speed ratios, and their alignment with traffic behavior. It is important to note, however, that Cluster numbers (0, 1, 2) in K-Means do not carry intrinsic meaning and were interpreted postscript based on the average jam factor and speed ratio within each group. A revision to this project wouldn't rely on index position but could map labels by cluster ID to a congestion score. As the model currently stands, the K-Means clustering method output 3 groupings, where cluster 0 = "Congested", cluster 1 = "Moderate Flow", and cluster 2 = "Free Flowing" (Table 2).

Cluster	Segment Count	Avg Jam Factor	Avg Speed Ratio	Most Common Highway
0	100	3.724000	0.394940	Secondary
1	116	2.281897	0.618021	tertiary
2	143	0.560140	0.831127	secondary

TABLE 2: K-MEANS OUTPUT (5PM PEAK PERIOD)

DATA AGGREGATION AND SUMMARIZATION

Post-clustering, with each group successfully labeled based on its statistical properties, the labels were assigned to the original segments for further analysis. Segments were first visualized in a map format using folium with standardized colors (red, orange, green) to denote congested, moderate, and free flowing roads (Image 1). Raw data for each segment, including attribution for weighted features, was exported to a CSV for use in the prompt engineering phase. The output includes the road name, highway classification, Jam Factor, Speed Ratio, Congestion Label, and the K-Means cluster number.

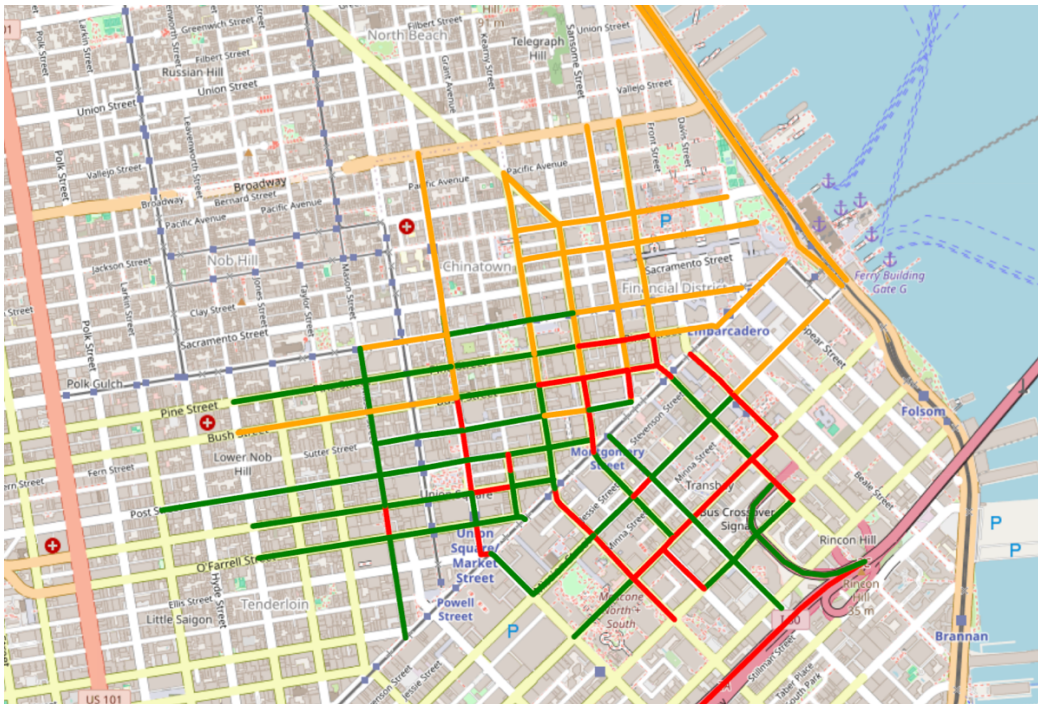


IMAGE 1: PEAK PERIOD CLUSTERING MAP

Clustered results were then aggregated along two key parameters: road name, to determine which individual streets had high or low congestion; and highway type, to assess patterns of congestion across different road classes. These results were output as separate CSV files for data ingestion. The tables ultimately informed comparative analysis between datasets and contributed to the formulation of insights for transportation policy.

PROMPT ENGINEERING

The final phase of the project utilizes large language models (LLMs), specifically in this project identified as Chat-GPT-4, to interpret the spatial and clustering results output from live traffic data. By creating structured prompts and supplying a labelled and categorized dataset from the machine learning model, an LLM was used to compare results from peak and off-peak periods, infer spatial data, and make policy suggestions to improve roadway conditions.

Two sets of output data were generated for analysis by the LLM, separated by temporal range:

1. Peak Period: Monday at 5:00 PM PST (expected heavier congestion)
2. Off-Peak Period: Sunday at 5:00 PM PST (expected lighter congestion)

The files prepared for Chat-GPT ingestion included the following:

1. *highway_congestion_summary.csv*: A summary of congestions statistics by road classification, including Segment Count, average Jam Factor, average Speed Ratio, and most common congestion label.
2. *road_congestion_summary.csv*: Aggregated statistics for individual road names, including the same weighted features in the above file.
3. *simplified_congestion.csv*: Segment-level datasets data sets with field corresponding to Road Name, Highway, Jam Factor, Speed Ratio, Congestion Label, and Cluster Number.
4. *Folium Map Screenshots*: Screenshots were taken for both time periods, visualizing segment congestion levels by congestion label (Congested, Moderate Flow, Free Flowing).

These visual and tabular data inputs were provided to the language model to conduct spatial interpretation and ultimately prescribe policy suggestions for improvement to the congested areas.

Next, a prompt engineering strategy was designed to explore several objectives. The goal of the project was to use structured prompts to summarize spatial patterns from model outputs and suggest urban design improvements. We would then compare two to three prompt variants to observe differences in quality of output by the model. A descriptive and prescriptive template was developed:

1. *Temporal Comparison (Descriptive)*
 - a. "Based on these two congestion snapshots from Sunday and Monday, which roads experienced the largest decrease in congestion?" Which roads remained persistently congested?"
2. *Policy and Planning Implications (Prescriptive)*
 - a. What recommendations would you make to the city based on these traffic patterns? Which roads experience the most congestion, and should be prioritized for infrastructure or policy improvements?"

Prompts were refined to result in clear, grounded insights, and screenshots were attached with the tabular data where visual confirmation was needed. They were executed within the ChatGPT interface with the model loaded using the structured tables, an image upload for map screenshots, and instructions where needed. Responses were then analyzed for alignment with observed trends, accuracy in data comparison, and the value of improvement recommendations.

RESULTS & ANALYSIS

TRAFFIC CLUSTERING RESULTS

Using real-time traffic flow data accessed from the HERE Traffic API and enriching road segments with OpenStreetMap road classifications, K-Means clustering was utilized to locate congestion patterns in downtown San Francisco during two time periods: Monday at 5:00PM and Sunday at 5:00PM, considered peak and off-peak periods sequentially. Clusters were generated and labelled based on a congestion score primarily derived from the Jam Factor and Speed Ratio attributions.

The table below denotes the Monday peak period summary of congestion by highway type. The model found that Secondary roads were the most frequently congested, with a moderate jam factor of 2.11 and a speed ratio of 0.62. Tertiary/Tertiary Link roads had the highest jam factor at 5.1 and the lowest speed ratio, although these segments were only labelled as “Moderate Flow”. Busways and Living Streets displayed the lowest congestion rates, with a jam factor of 1.39 and a speed ratio of 0.72.

highway	segment_count	avg_jamFactor	avg_speed_ratio	most_common_congestion
tertiary, tertiary_link	2	5.1	0.29705882	Moderate Flow
unclassified	4	3.025	0.58703013	Free Flowing
tertiary_link	2	2.3	0.58059208	Free Flowing
secondary	116	2.10948276	0.62470277	Congested
secondary_link	8	2	0.63408631	Moderate Flow
residential	56	1.97678571	0.63530912	Congested
tertiary	156	1.92820513	0.65047026	Free Flowing
busway	13	1.39230769	0.72421396	Congested
living_street	2	0	0.96428571	Congested

TABLE 1: MONDAY HIGHWAY TYPE SUMMARY

Data output for the Sunday, or off-peak, period can be found in Table 2 below. During this timeframe, Tertiary/Tertiary Link roads again had the highest jam factor of 8.4 with the lowest speed ratio of 0.29, which tells us that these streets were highly congested on both weekdays and weekends. Secondary roads had a similar moderate jam factor (2.5) and speed ratio (0.62) to Monday and remained “Congested”. This suggests that there is consistent traffic on these roads regardless of the day. Residential and Busway segments had moderate flow but were labelled as “Congested”, even with speed factors above 0.66, which may require a reevaluation of the clustering parameters.

highway	segment_count	avg_jamFactor	avg_speed_ratio	most_common_congestion
tertiary, tertiary_link	2	8.4	0.29705882	Congested
unclassified	4	3.275	0.55529722	Moderate Flow
secondary	116	2.50344828	0.62284075	Congested
busway	13	2.11538462	0.66740516	Congested
residential	56	2.03392857	0.66279364	Congested
secondary_link	8	1.9625	0.66251513	Congested

tertiary	156	1.89871795	0.70176896	Moderate Flow
tertiary_link	2	1.55	0.7368421	Moderate Flow
living_street	2	1	0.7455357	Congested

TABLE 2: SUNDAY HIGHWAY TYPE SUMMARY

Key observations between from Monday to Sunday travel patterns are recorded in Table 3. The results from a highway classification-based summarization are mixed. Tertiary roads experienced higher congestion off-peak, which could be explained by localized conditions or by lower vehicular drivers on the Cinco de Mayo holiday. Secondary roads were overall consistently congested regardless of day, and Busways had a slight improvement in traffic conditions from Sunday to Monday. Residential roads overall were less congested over the weekend.

Road Type	Jam Factor Change	Speed Ratio Change	Comment
Tertiary	↑ (5.1 → 8.4)	↓ (0.30 → 0.30)	More congested off-peak, possibly localized
Secondary	↑ (2.11 → 2.50)	≈ (0.62 → 0.62)	Consistently pressured
Busway	↓ (2.57 → 2.12)	↑ (0.61 → 0.67)	Slight improvement
Residential	↓ (3.08 → 2.03)	↑ (0.63 → 0.66)	Much less congested on Sunday

TABLE 3: MONDAY VS. SUNDAY HIGHWAY TYPE SUMMARY

The tables below display the road-level summaries for Monday (Table 4) and Sunday (Table 5) by most congested streets. At the segment level, 4th St/5th St demonstrated persistent congestion across both time periods, with jam factors over 8.50 and speed ratios under 0.20. Roads like Bush/Market St and Pine St showed significant congestion during the weekend with relief on Monday, which suggests a possible localized traffic surge due to events downtown. These findings show the variance in temporal data and emphasize the fact that transportation planning methods may be required for these corridors regardless of assumed peak day travel patterns.

Road Name	Avg Jam Factor	Avg Speed Ratio	Congestion Label(s)
4th St/5th St	8.80	0.21	Moderate Flow (<i>likely mislabeled</i>)
Bush St	3.77	0.48	Mixed - Congested + Free Flow + Moderate
Sansome St	3.63	0.36	Free Flowing + Moderate
4th St	3.30	0.51	Congested + Moderate
Howard St	3.29	0.53	Congested + Moderate

TABLE 4: MONDAY ROAD-LEVEL SUMMARY

Road Name	Avg Jam Factor	Avg Speed Ratio	Congestion Label(s)
Bush/Market St	9.20	0.17	Congested
4th St/5th St	8.50	0.26	Congested
Pine St	7.77	0.35	Congested

Mission St	4.55	0.45	Mixed: Congested, Free Flow, Moderate
Broadway	3.80	0.35	Moderate Flow

TABLE 5: SUNDAY ROAD-LEVEL SUMMARY

Based on the traffic patterns identified by the clustering algorithm, we can recommend several interventions for congestion improvement:

1. *Congested Corridors (4th/5th St, Bush/Market, Pine St)*
 - a. Dedicated bus or high occupancy vehicle lanes
 - b. Potential lane configurations or congestion pricing
2. *Secondary Arterials*
 - a. Adaptive signal timing to account for traffic fluctuations
 - b. Turn movement restrictions for intersections with high jam factors
3. *Off-peak Congestion*
 - a. Event-based signal timing plans for temporary traffic surges

These interpretations provide a reference for data-based urban transportation planning. The next section of the study applied prompt engineering to evaluate whether an LLM could identify similar patterns and generate meaningful policy interpretations unsupervised.

LLM-BASED INTERPRETATION

To produce planning insights, GPT-4 was used to analyze and summarize the clustering data outputs. Structured prompts were loaded with CSV tables and Folium map screenshots, as well as detailed instructions for analysis. The language model first identified road segments that showed the most improvement from the off-peak period on Sunday at 5pm to the peak period on Monday at 5pm (Table 1). These roads experienced significant congestion reduction on Monday comparative to off-peak hour, with the LLM identifying atypical Sunday travel behavior. Road segments that became more congested on Monday were also highlighted, which is consistent with typical peak hour commuting travel (Table 2).

Road Name	Jam Factor Change	Speed Ratio Change
Bush St/Market St	-6.10	+0.17
Pine St	-4.64	+0.13
Steuart St	-1.70	+0.16
Clay St	-1.60	+0.18
Mission St	-1.48	-0.04

TABLE 1: SUNDAY TO MONDAY TRAFFIC IMPROVEMENT

Road Name	Jam Factor Change	Speed Ratio Change
Bush St	+2.92	-0.29
Harrison St	+2.90	-0.53
Sansome St	+1.70	-0.22
Montgomery St/Columbus Ave	+1.50	-0.33
4th St	+1.30	-0.18

TABLE 2: SUNDAY TO MONDAY CONGESTION INCREASE

Other key findings from GPT-4 include:

- Persistent congestion was recognized on several key arterials (including Bush St, Harrison St, and Sansome St). Tertiary roads on Sunday had an average jam factor of 8.4, which dropped to 5.1 on Monday, showing improvement.
- The model highlighted inefficient traffic patterns on secondary roads that may not typically be prioritized for infrastructure improvements.
- Residential and busway segments remained congested on both days, with only minor fluctuation in average jam factors.
- Unclassified roads saw mild improvement in speed ratios and lower congestion on Monday.

The prompt engineering phase provided several intervention recommendations:

1. *Congested Corridors (Bush St, Harrison St, and Sansome St)*
 - a. Signal timing optimization is recommended for high-volume arterials where congestion persists no matter the day.
 - b. Consider a road diet or dedicated transit lanes where busways remain congested.
2. *Off-peak Congestion*
 - a. Routing and signage review may warrant better traffic control on these roads for event-based or weekend congestion. This specifically applies for the Bush St/Market St area.

Based on the prompt and input data, the language model has identified traffic trends, and suggested roads or regions that may benefit from targeted planning interventions.

COMPARATIVE INSIGHTS

The spatial clustering analysis revealed several congested corridors that defy typical peak vs non-peak traffic behavior and provided us with a background to make infrastructure improvement suggestions. A key finding from the data was that 4th/5th St experienced the highest congestion levels across both days with a lower speed ratio. This corridor functions as a critical connector to downtown, and this persistent traffic overload suggests that there is a failure in vehicular capacity. Bush/Market St was especially congested during the off-peak period, which may indicate there is a weekend-specific demand incurring high levels of traffic, such as special events or rideshare surges. The Secondary road type experienced moderate congestion overall, and due to the prevalence of this road type and its functional role in supporting local access as well as highway connection, they may benefit from infrastructure improvements as well. Tertiary and Residential roads were generally comparable and

moderately free flowing, although the clustering process identified Residential roads as “Congested” both days: the implication of this result could be that the clustering parameters need to be changed.

Much of this information was substantiated by the prompt-driven analysis conducted in ChatGPT. The LLM effectively interpreted CSV tables and visual maps to identify spatial trends and offered realistic recommendations with urban traffic planning practices. An interesting insight from the data, where congestion rates were higher on some streets due to anomalous Sunday traffic, was immediately recognized and addressed in the intervention planning stage. Overall, it seems that the LLM-based analysis focused heavily on the changes in volumes between days, and less so on the overall congestion patterns by roadway type. An ideal approach would be to take these findings and synthesize them with the results and suggestions derived solely from the clustering data results.

CONCLUSION

The goal of this project was to explore the research question, **can prompt engineering and lightweight AI models help detect and interpret traffic congestion patterns in urban areas?** Using geospatial AI tools, unsupervised machine learning models, and large language models, we demonstrated that prompt engineering can support congestion analysis when paired with labelled spatial data and clustering models.

Real-time traffic flow data was ingested from the HERE API and enriched with attribution from OpenStreetMap, then clustered using K-Means defined by features such as Jam Factor and Speed Ratio. This method allowed for successful grouping of the data based on actual traffic conditions to label roadways as “Congested”, “Moderate Flow”, and “Free Flowing”. The results of the clustering algorithm were compared over two timeframes to highlight the variability of congestion rates based on peak travel periods. These findings were interpreted through prompt engineering with Chat-GPT-4, which utilized tabular data and map visualizations to recognize traffic trends, weigh congestion levels across time and road types, and offer intervention recommendations. This phase ultimately confirmed the usefulness of LLMs not as a replacement for spatial analysis, but as a tool for interpreting and collaborating on results.

In conclusion, this project validates the question of whether prompt engineering, when combined with real-time spatial data and machine learning processes, can offer a leg up in understanding urban traffic congestion patterns. Through these methods, we were able to examine live data, derive conclusions from machine learning models, and produce real-world strategies to help reduce congestion at key locations in an urban environment.

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